

Initial Load

```
passengers_day1 = [  
  
    (101, "Rahul Sharma", "Hyderabad", "Economy", "India"),  
    (102, "Priya Reddy", "Bangalore", "Business", "India"),  
    (103, "Amit Kumar", "Mumbai", "Economy", "India"),  
    (104, "Sneha Patel", "Delhi", "Premium Economy", "India"),  
    (105, "Farhan Ali", "Chennai", "Economy", "India")  
  
]  
  
columns = [  
    "passenger_id",  
    "passenger_name",  
    "city",  
    "travel_class",  
    "country"  
]  
  
df_day1 = spark.createDataFrame(  
    passengers_day1,  
    columns  
)
```

Incremental Load

```
passengers_day2 = [  
  
    (102, "Priya Reddy", "Bangalore", "First Class", "India"),  
    (104, "Sneha Patel", "Hyderabad", "Premium Economy", "India"),  
    (106, "Neha Singh", "Pune", "Economy", "India"),  
    (107, "Arjun Verma", "Kochi", "Business", "India")  
  
]
```

```
df_day2 = spark.createDataFrame(  
    passengers_day2,  
    columns  
)
```

SCD Type 1 Scenario

Before:

passenger_id	city	travel_class
102	Bangalore	Business

After Merge:

passenger_id	city	travel_class
102	Bangalore	First Class

Old value:

```
Business
```

is overwritten.

Exercises

Create Delta

1. Create passengers_delta table.
2. Load Day1 data.
3. Verify record count.
4. Read Delta table.
5. View Delta history.

Merge / Upsert

6. Merge Day2 data.
7. Update existing passengers.
8. Insert new passengers.
9. Verify passenger 102 travel class change.
10. Verify passenger 106 insertion.

Time Travel

11. Read Version 0.
12. Read latest version.
13. Compare Version 0 and latest version.
14. Check passenger 102 in both versions.
15. Check passenger 104 in both versions.

Maintenance

16. Run OPTIMIZE.
17. Run ZORDER BY(city).
18. Delete passenger 105.
19. View Delta history again.
20. Run VACUUM.